Supplementary Information: Towards welfare-oriented recommendations in activity-travel behavior

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1 Proof of Proposition 1

Proof. By definition, the non-abstention probability $p(\theta)$ is continuous and strictly decreasing in θ, bounded by $p(\theta_{\min}) = 1$ (unconstrained recommendation) and $p(\theta_{\max}) = 0$ (strict filtering precluding all candidates). Because the filter progressively eliminates the lowest-scoring candidates, the conditional expected utility of a surviving recommendation, $\mu(\theta)$, is non-decreasing in θ.

Differentiating $\bar{U}(\theta)$ with respect to θ yields:

$$\bar{U}'(\theta) = p(\theta)\mu'(\theta) + (\mu(\theta) - \bar{U}^{\text{org}})p'(\theta), \tag{1}$$

where $p(\theta)\mu'(\theta)$ is strictly non-negative and represents the marginal welfare gain from filtering out sub-optimal options and $(\mu(\theta) - \bar{U}^{\text{org}})p'(\theta)$ represents the marginal penalty of increased abstention. It is non-positive because $p'(\theta) \leq 0$ and $\mu(\theta) \geq \mu(\theta_{\min}) > \bar{U}^{\text{org}}$.

Evaluating the derivative at the boundaries reveals a change in sign. At minimum strictness, $p(\theta_{\min}) = 1$ and $p'(\theta_{\min})$ approaches zero, leaving $\bar{U}'(\theta_{\min}) > 0$. The RS strictly improves upon organic choice. As $\theta \to \theta_{\max}$, universal abstention occurs ($p(\theta_{\max}) \to 0$), causing $\bar{U}(\theta)$ to converge to the organic baseline $\bar{U}^{\text{org}}$ from above, implying a negative trajectory.

Because $\bar{U}'(\theta)$ is continuous and changes sign from positive to negative, the Intermediate Value Theorem guarantees the existence of at least one interior critical point $\theta^* \in (\theta_{\min}, \theta_{\max})$ where $\bar{U}'(\theta^*) = 0$.

Finally, because the residual uncertainty ϵ_{ik} in the utility model is normally distributed, its probability density function is strictly log-concave. This ensures a monotonically increasing hazard rate for the welfare-score distribution, guaranteeing that the derivative $\bar{U}'(\theta)$ crosses zero exactly once. Therefore, the critical point θ^* is the unique global maximizer of expected user welfare. □

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2 Simulation Details

2.1 Recommender Stack Details

The simulation implements a multi-platform recommender stack that mirrors the heterogeneous RS landscape users encounter in practice. Two platform types are instantiated, each producing a base score B_k that is then blended with personalization via Eq. 11. We check the robustness of all parametric assumptions (detailed below) in §4.

2.1.1 Google Maps Replica. This platform computes the base score B_k for candidate place k as a convex combination of three normalized signals:

$$B_k = w_{\text{prom}} \cdot \text{Prom}_k + w_{\text{rel}} \cdot \text{Rel}_k + w_{\text{prox}} \cdot \text{Prox}_k \tag{2}$$

where the weights $w_{\text{prom}} = 0.45$, $w_{\text{rel}} = 0.35$, and $w_{\text{prox}} = 0.20$ are normalized to sum to one. *Prominence* (Prom) combines the place's average rating (scaled to $[0, 1]$) and the log-transformed review count (min-max normalized), weighted 0.55 and 0.45 respectively. *Relevance* (Rel) is a Jaccard similarity between the place's keyword tags and the user's query keywords (75% weight) plus interest keywords (25% weight). *Proximity* (Prox) applies exponential distance decay: $\text{Prox}_k = e^{-d_{ik}/d_0}$, where d_{ik} is the Euclidean grid distance scaled to kilometers and $d_0 = 5$ km. The final score blends this base with personalization: $\hat{V}_{ik}^{\text{act}} = (1 - \lambda)\, B_k + \lambda\, P_{ik}$, with default $\lambda = 0.10$ (clamped to $[0, 0.35]$).

2.1.2 Popularity-Based Recommender. This platform computes B_k from three signals: average rating (scaled to $[0, 1]$, weight 0.25), log-transformed review count (weight 0.35), and a general popularity signal based on total visit count (weight 0.40). Raw scores are min-max normalized before weighting. The default personalization weight is $\lambda = 0.12$ (also clamped to $[0, 0.35]$). This platform does not use proximity or relevance signals; it represents platforms like OpenTable or Ticketmaster where popularity and social proof dominate ranking [5, 6].

2.1.3 Subtype Routing. We route each leisure subtype to the appropriate platform. Table 1 shows the mapping. When multiple platforms serve a subtype (e.g., food), recommendations from each platform are returned separately and the agent selects from the merged set. This set of RS are selected from popular apps/companies operating in these domains within the continental United States in 2026.

2.1.4 Personalization Learning. Both platform types share the same personalization mechanism (Eq. 12), in which we use $\Omega = 0.6$ to give higher weight to place affinity (motivated by recency bias in

Table 1: Leisure subtype to recommender platform routing.

Leisure subtype	Platform(s)
Food (dine-in)	Google Maps replica + Popularity RS (i.e., OpenTable)
Food (takeout)	Google Maps replica + Popularity RS (i.e., OpenTable)
Live music	Popularity RS (i.e., Spotify)
Workout / fitness	Popularity RS (i.e., ClassPass)
Café	Google Maps replica
Museum	Google Maps replica
Park	Google Maps replica

human mobility [1]. When a user completes an activity and provides feedback, place-level and keyword-level affinities are updated incrementally (Eq. 13) with a learning rate of $\gamma = 0.12$. Negative feedback applies a stronger update ($-\gamma$) than positive feedback ($+\gamma$), and keyword affinities update at 70% of the place-level rate. Over the course of the simulation, this causes the personalization score P_{ik} to drift toward places and content types that the agent has consistently enjoyed, and away from those that produced negative experiences.

2.1.5 Computing PUP and RM. Given estimates $\hat{V}_{ik}$ and $\hat{C}_{ik}$, the RS computes the welfare criteria as follows.

For PUP, the RS estimates the probability that net utility is non-negative. Assuming the residual uncertainty $\epsilon_{ik} \sim \mathcal{N}(0, \sigma_i^2)$ with variance σ_i^2 learned from prediction errors in the feedback history:

$$\Pr[U_{ik} \geq 0] \approx \Phi\left(\frac{\hat{V}_{ik} - \hat{C}_{ik}}{\sigma_i}\right) \tag{3}$$

where $\Phi(\cdot)$ is the standard normal CDF. The recommendation passes PUP at threshold α if this probability exceeds α.

For RM, the RS estimates expected regret by comparing the recommended activity against the set of candidates it would have surfaced in a standard ranking. Let $\hat{U}_{ij} = \hat{V}_{ij} - \hat{C}_{ij}$ for each candidate j. Then:

$$\hat{R}_{ik} = \max_{j \in \mathcal{A}} [\hat{U}_{ij}] - \hat{U}_{ik} \tag{4}$$

The recommendation passes RM at tolerance ϵ if $\hat{R}_{ik} \leq \epsilon$. In practice, the RS can compute this efficiently over its top-N candidates rather than the full activity set.

2.2 Spatial Environment

The city is a $G \times G$ grid (default $G = 18$) where each cell is assigned a zone type—residential, employment, leisure, or mixed—according to configurable probabilities. Block spacing determines the spatial scale (default 0.5 km per cell). Distance is computed as Euclidean distance scaled to kilometers.

Each zone produces a configurable number of places drawn from category distributions appropriate to its zone type. Leisure zones produce museums, parks, live music venues, and cafés; mixed zones produce a broader category mix; residential zones produce cafés, takeout, parks, and fitness venues. Each place is assigned a random rating (3.5–4.9), review count (20–5000), popularity score, and a keyword tuple. A required-category check ensures that all seven leisure subtypes have at least one matching POI.

2.3 Travel Cost Estimator

The Travel Cost Estimator (TCE) learns each user's value of time $\hat{v}_i$ from continuous feedback via an exponential moving average (EMA) with learning rate $\eta_{\text{TCE}} = 0.15$. After each trip, the TCE observes the user's continuous satisfaction signal $s_{ik} \in [-1, 1]$ alongside the realized travel time and monetary cost. If the user reports dissatisfaction with a distant activity despite high preference match, the VOT estimate is adjusted upward by 5%; if the user reports satisfaction with a longer trip, VOT is adjusted downward by 3%. The asymmetry between these adjustments reflects the well-documented principle that negative experiences carry greater informational and psychological weight than comparably-sized positive experiences [2, 4, 8]. Dissatisfaction with a time-costly trip is a strong signal that the traveler's marginal disutility of travel time exceeds the current estimate, whereas satisfaction with a long trip is a weaker signal because it may reflect activity enjoyment rather than a genuinely low time cost. The resulting $5:3$ ratio lies within the range of empirically observed loss-aversion coefficients (≈ 1.5–2.5) reported in riskless-choice experiments [8], and it prevents the failure mode in which streams of mildly positive feedback on short trips cause the VOT estimate to collapse toward zero.

Estimates are bounded between \$5/hr and \$50/hr. The TCE also maintains a per-user uncertainty estimate $\sigma_i = \max(0.15, 1/\sqrt{n_i})$ that decreases with the number of observations n_i, and is used in PUP computation via Eq. 3.

2.4 Willingness-to-Accept Model

The dynamic willingness-to-accept parameter η_i is computed as follows:

$$\eta_i = \text{clamp}\Big(\eta_i^{\text{base}} + \Delta^{\text{qual}} + \Delta^{\text{mem}},\ 0.05,\ 0.95\Big), \tag{5}$$

where η_i^{base} is a baseline willingness calibrated from the agent's trust in platforms and autonomy preference, Δ^{qual} scales linearly with the RS's own score for the suggested place, and Δ^{mem} increases with the count of previously accepted recommendations. Concretely,

$$\begin{aligned}
\eta_i^{\text{base}} &= 0.52 + 0.40\,(T_i - 0.5) - 0.40\,(A_i - 0.5),\\
\Delta^{\text{qual}} &= 0.18\,(S_k - 0.5),\\
\Delta^{\text{mem}} &= 0.015 \cdot \min(5,\ n_{\text{accepted}}),
\end{aligned}$$

where T_i and A_i are the agent's trust-in-platforms and autonomy-preference latent variables (both $\in [0, 1]$), S_k is the normalised RS score of the suggested place, and n_{accepted} is the agent's cumulative count of accepted recommendations (capped at five). The clamp bounds $[0.05,\ 0.95]$ preserve a floor of exploration and a ceiling of skepticism.

3 Agent Model

This appendix collects implementation details for the agents themselves: the cross-trait profile generator that produces each agent's demographics (§3.1), the value of time calibration process (§3.2), and agents' psychological traits (§3.3).

3.1 Synthetic Population Generation

We construct a synthetic population using an orthogonal-factor design that minimizes cross-factor correlations [3]. This allows clear attribution of variation in recommendations to specific persona attributes. Each synthetic persona is defined over 18 factors relevant to local leisure decisions, including sociodemographics (age, biological sex, income), home location (e.g., urban, suburban, rural), mobility resources (car/transit access), budget, time window preference, group composition (e.g., solo, with friends, with kids), among others. Each factor has 2-5 discrete levels, yielding a large combinatorial space from which we sample 250 highly diverse personas. Table 2 summarizes the full factor design used to generate our synthetic personas.

3.2 Value of time calibration

Each agent's value of time is computed as $v_i = \gamma_w \cdot (\text{income}_i / H)$, where income_i is the agent's annual income, $H = 2080$ is the assumed annual work hours, and $\gamma_w = 0.5$ is a scaling factor reflecting that leisure-trip travel time is typically valued at a fraction of the market wage rate. This choice is grounded in a large transportation-economics literature: Small's [7] review of empirical VOT estimates finds that the mean value of time for commuting and personal travel clusters around 50% of the gross wage, and the U.S. Department of Transportation's guidance for benefit-cost analysis explicitly recommends valuing personal local travel at 50% of the hourly median household income [9]. Yan et al. [11] further show that LLM-based synthetic travelers reproduce this ≈50% VOT benchmark across a range of income and trip-purpose contexts. For an agent earning \$60,000/year, the resulting $v_i \approx \$14.4$/hr is therefore consistent with both empirical estimates and official practice. The parameter γ_w can be adjusted to reflect different assumptions about the marginal utility of leisure time.

3.3 Psychological trait model

Each agent carries a Big Five personality vector (openness, conscientiousness, extraversion, agreeableness, neuroticism) and four latent variables: maximization tendency, trust in platforms, autonomy preference, and algorithmic awareness.

The latent traits modulate ten behavioral coefficients via a parameterized mapping: platform affinity, autonomy guard, planning orientation, social orientation, variety seeking, risk aversion, budget sensitivity, trend susceptibility, AI affinity, and feedback loop strength. Each coefficient is computed as a weighted sum of the relevant personality traits and latent variables, following the integrated choice and latent variable (ICLV) approach [10], which embeds psychometric indicators inside discrete-choice models so that unobserved attitudinal constructs can influence behavioral parameters in a theoretically grounded way. The behavioral coefficients, in turn, shift the agent's operational decision parameters: preference weights, attitude strengths, loss aversion, satisficing threshold, and the dynamic willingness to accept recommendations (η).

4 Robustness Checks

To assess how sensitive the paper's main conclusions are to parameter choices, we run a set of one-factor-at-a-time (OFAT) sensitivity sweeps grouped into three tasks. Set 1 varies knobs that govern the welfare-criteria uncertainty state (the belief standard deviation σ_i used inside PUP and the TCE that learns each user's value of time). Set 2 varies knobs that govern how the base recommender stack learns from user feedback (personalization weights, keyword discount, feedback-strength clamps). Set 3 varies knobs that shape the base stack itself (GMaps prominence/relevance/proximity weights) and the synthetic place catalog it reads from (rating, review-count, popularity-multiplier distributions). Across the three tasks we run 35 single-knob perturbations.

Each configuration reuses the paper's full-scale simulation with a matched-seed paired-comparison design across five random seeds. For each configuration we re-run every treatment whose output the perturbed knob can affect. Reported cell means are averages over the five seeds; reported standard deviations are between-seed SDs. Paired-comparison metrics (% harmed, Over-Recommendation Cost) are computed per seed against the matched No RS counterfactual at the baseline configuration and averaged across seeds; absolute statistics (mean utility, negative-utility rate, Gini) are per-run statistics averaged across seeds directly.

Across the 35 single-knob perturbations, the paper's core ordering (No RS $<$ Standard RS $<$ PUP $\approx$ RM $<$ Oracle) is preserved in 34 cases. The single exception is Set 1's *K2-hi* (uncertainty-decay numerator set to 2.0, i.e. σ inflated by a factor of two): RM's mean per-trip utility collapses to +0.0646, *below* Standard RS's +0.0831, and its percent-harmed rate rises to 38.96% vs Standard RS's 38.0%. This is RM's known failure mode: the ε-regret band compares a point estimate of regret against a fixed tolerance, so doubling σ shifts many beneficial recommendations into the reject region. PUP-0.3 under the same perturbation still reads +0.2284 utility and 31.4% harmed because its probability-of-positive framing absorbs the larger σ through the normal CDF rather than against a hard threshold. The remaining 34 perturbations leave both welfare-constrained policies strictly above Standard RS in mean utility.

Set 1 (uncertainty / TCE, Figure 1). Two knobs dominate: K2 σ-inflation is the only regime-change perturbation, and K5 VOT clamp widening costs PUP about 0.04 utility by letting the empirical-Bayes implied VOT climb beyond what is reasonable, which in turn inflates the `estimate_travel_cost` prediction and over-filters the trips PUP is meant to evaluate.

Set 2 (feedback learning, Figure 2). P2 (personalization weight) is the dominant knob: doubling it to $(0.20, 0.24)$ costs PUP 0.033 utility and leaves RM nearly unchanged, while halving it lifts both. This is the Task-2 counterpart of PUP's σ-sensitivity in Set 1: here it is PUP that degrades faster than RM as the personalization signal is amplified, because PUP is the less-restrictive gate and is therefore more exposed to the larger shortlist churn.

Set 3 (RS stack / catalog, Figure 3). GMaps proximity scale is the dominant knob: tightening to 2.0 km lifts RM to +0.2611 (the highest RM cell in the grid), while loosening to 10.0 km drops RM to +0.1950 (the lowest); PUP is nearly flat across both levels. This is the Task-3 complement to Set 2's P2 effect: here RM is the more exposed gate because the ε-band assumes the base ranking is well-aligned with realized utility, whereas PUP's probability-of-positive gate is less sensitive to ranking quality. S1 (GMaps prom/rel split)

Table 2: Persona factor design and number of levels.

Factor	Options	Levels
Age	{teen, young adult, mid adult, old adult, senior}	5
Biological Sex	{male, female}	2
Income	{ <$35k, $35-75k, $75k-125k, $125k+ }	4
Home location	{urban core, inner suburb, outer suburb, rural fringe}	4
Car access	{own car, carshare, no car}	3
Transit access	{high, medium, low}	3
Budget (for an outing)	{low, medium, high}	3

and S3 (Popularity weights) produce smaller asymmetric shifts in the same direction. Standard RS remains uniformly below both welfare-constrained policies across the entire grid.

References

[1] Barbosa, H., de Lima-Neto, F. B., Evsukoff, A., and Menezes, R. The effect of recency to human mobility. *EPJ Data Science 4*, 1 (2015), 21.

[2] Baumeister, R. F., Bratslavsky, E., Finkenauer, C., and Vohs, K. D. Bad is stronger than good. *Review of general psychology 5*, 4 (2001), 323–370.

[3] Box, G. E. P., Hunter, J. S., and Hunter, W. G. *Statistics for Experimenters: Design, Innovation, and Discovery.* John Wiley & Sons, May 2005. Google-Books-ID: bZ9bEQAAQBAJ.

[4] Kahneman, D., and Tversky, A. Prospect Theory: An Analysis of Decision under Risk. *Econometrica 47*, 2 (1979), 263–291.

[5] Kuo, W.-T., Wang, Y.-C., Tsai, R. T.-H., and Hsu, J. Y.-j. Contextual restaurant recommendation utilizing implicit feedback. In *2015 24th Wireless and Optical Communication Conference (WOCC)* (2015), IEEE, pp. 170–174.

[6] Nilashi, M., Ibrahim, O., Yadegaridehkordi, E., Samad, S., Akbari, E., and Alizadeh, A. Travelers decision making using online review in social network sites: A case on tripadvisor. *Journal of computational science 28* (2018), 168–179.

[7] Small, K. A. Valuation of travel time. *Economics of transportation 1*, 1-2 (2012), 2–14.

[8] Tversky, A., and Kahneman, D. Loss aversion in riskless choice: A reference-dependent model. *The quarterly journal of economics 106*, 4 (1991), 1039–1061.

[9] U.S. Department of Transportation. Revised Departmental Guidance on Valuation of Travel Time in Economic Analysis. Tech. rep., U.S. Department of Transportation, Office of the Secretary of Transportation, 2016.

[10] Vij, A., and Walker, J. L. How, when and why integrated choice and latent variable models are latently useful. *Transportation Research Part B: Methodological 90* (Aug. 2016), 192–217.

[11] Yan, Y., Liu, T., and Yin, Y. Valuing Time in Silicon: Can Large Language Model Replicate Human Value of Travel Time, July 2025. arXiv:2507.22244 [econ].

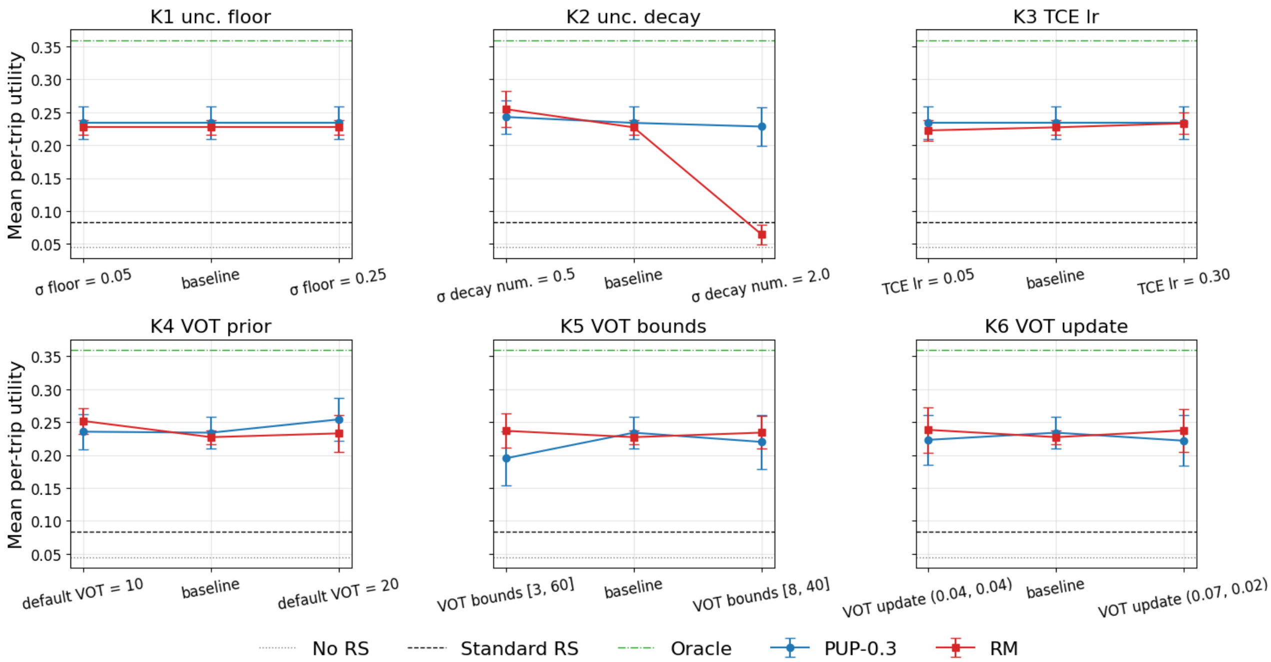

Figure 1: Set 1 OFAT sensitivity over uncertainty / TCE knobs (K1–K6). Each panel shows mean per-trip utility ± between-seed SD for PUP-0.3 (blue) and RM (ε=0.27, red) at the low, baseline, and high levels of one knob. Dotted grey line marks the No-RS reference; dashed green line marks the Oracle reference (both at baseline). Standard RS sits near 0.08 throughout and is shown as a grey series. The K2-hi cell is the only regime-change perturbation across all three tasks.

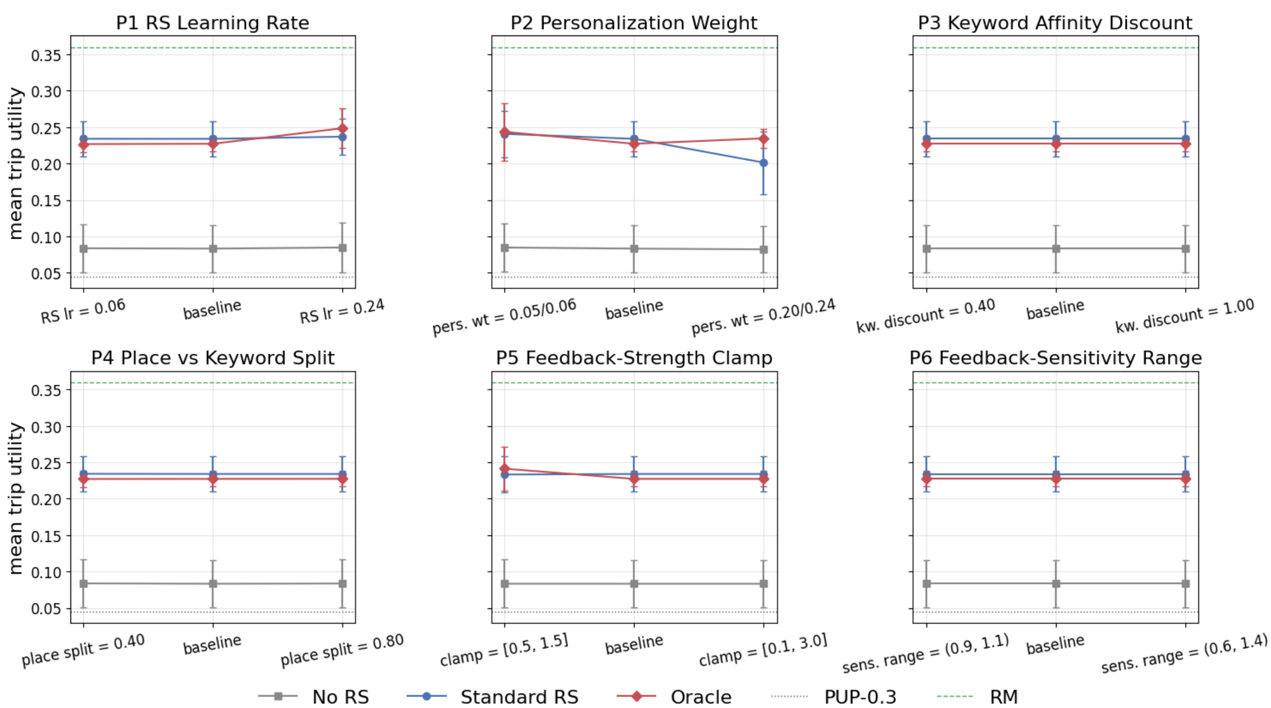

Figure 2: Set 2 OFAT sensitivity over personalization / feedback-learning knobs (P1–P6). P2 (personalization weight) is the dominant knob and is asymmetric: doubling it costs PUP-0.3 about 0.03 utility while leaving RM essentially flat.

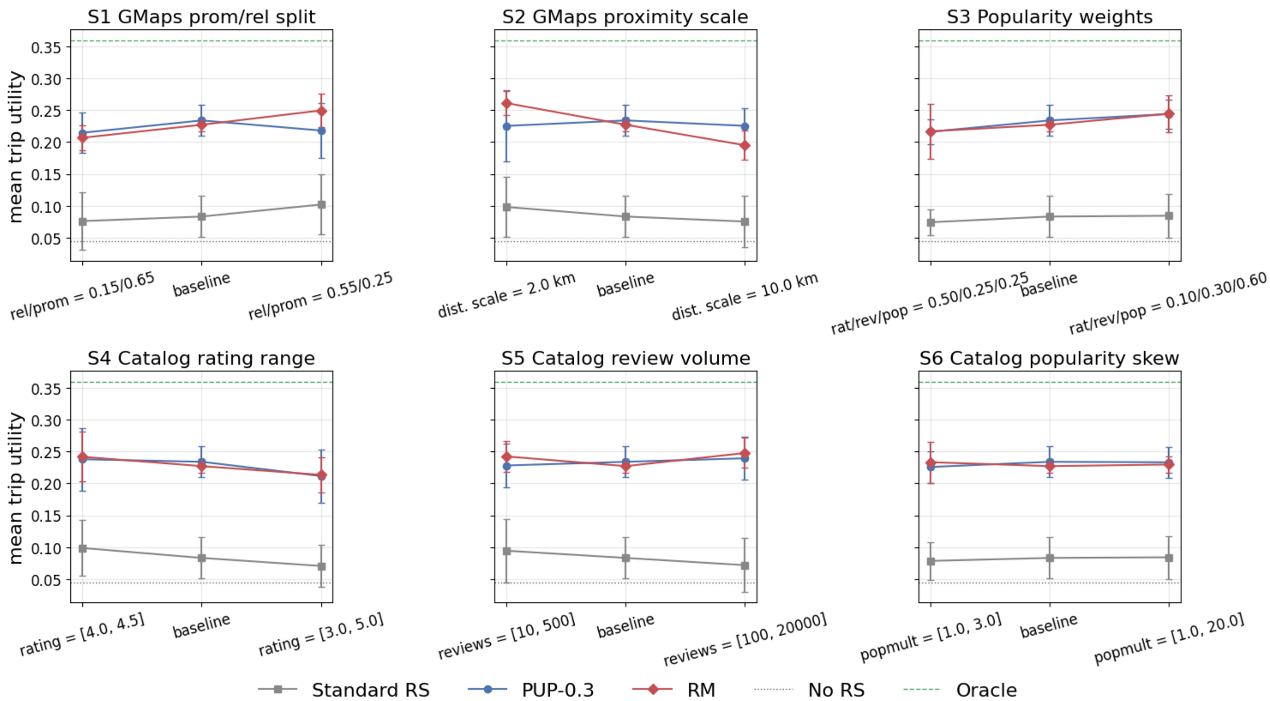

Figure 3: Set 3 OFAT sensitivity over RS-stack / catalog knobs (S1–S6). S2 (GMaps proximity scale) is the dominant knob, asymmetric in the opposite direction from Set 2: RM is the exposed policy under base-stack/geography misalignment, whereas PUP stays stable across both S2 levels.